2	**Ans: A** Before tying the 200 g mass, the rod is balanced, hence the CG of the rod is at X. After tying the 200 g mass and shifting the rod to balance again, the new CG of the rod is at 0.27 m from point X. by principle of moments: mass of rod × g × 0.27 = 0.200 × g × (1.0 – 0.10 – 0.27) mass of rod = 0.503 kg